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Unit-1-slides - Computational physics lecture notes

Computational Physics (Vellore Institute of Technology Bhopal University)

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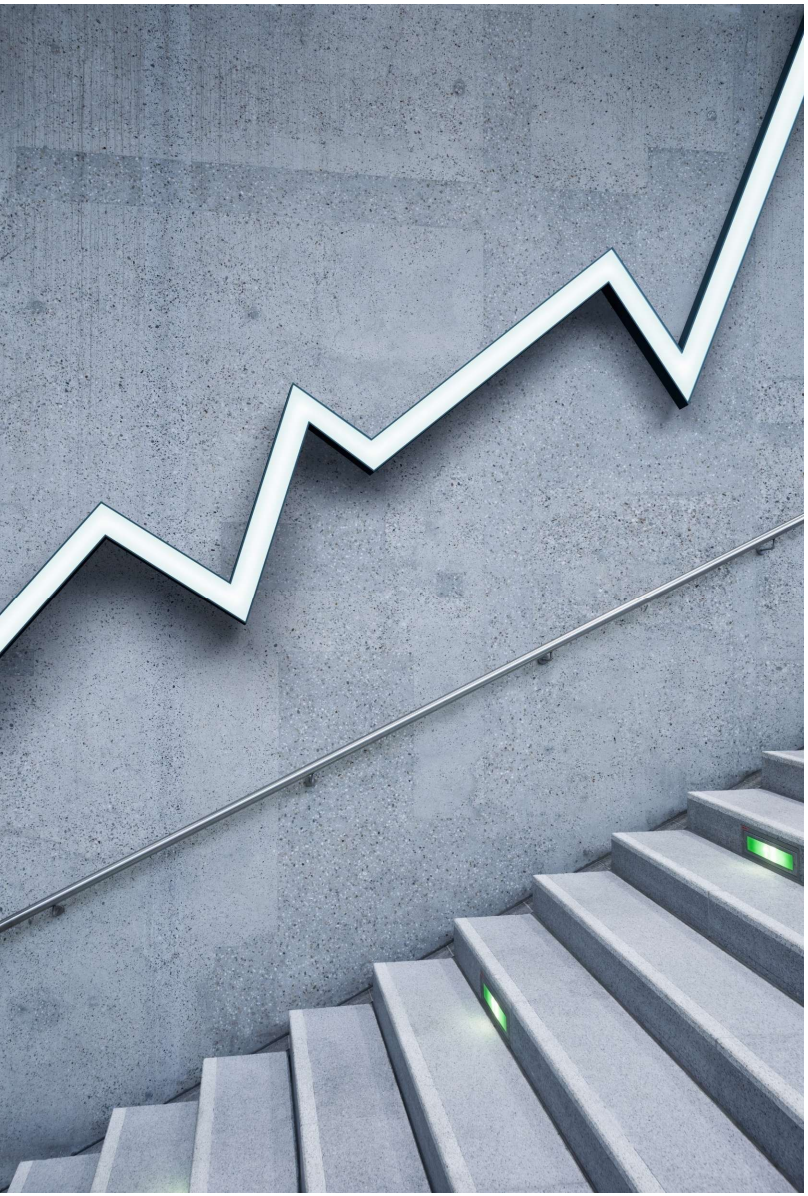

Computational Physics (Phy1003)

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Mod. No.	Module Description	Lect. Hrs
1	**Basic data operations** Fundamentals of computations, Data visualization in 2 and 3-dimension, Plotting and data interpretation in 2 and 3 dimension, data interpolation and fitting	5
2	**Mathematical Physics I** Scalars, Vector, Col linear vectors, Addition and subtraction of vectors, Scalar products, Vector products, Vector triple product, Gradient, Divergence, Curl, Matrix methods, Matrix operations, Arrays	8
3	**Mathematical Physics II** Sequential and random numbers, Concept of randomness, Measures of central tendency and dispersion, Frequency distribution, Error analysis, Root finding, Simple functions, Derivatives and integrals, Forward and backward difference, Central difference, Ordinary differential equations, Partial diffrential eqations	10

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4	**Particle dynamics; a computational approach I** Newton's equation of motions, Work energy theorem, conservative forces, Linear motion, non-linear motions, Uniform and non uniform motion, Collision of particles, Elastic collision, Inelastic collision, Conservation laws, Algorithm development and problem solving	8	1,2,3,4
5	**Particle dynamics; a computational approach II** Simple harmonic motion Simple harmonic oscillator : energy and phase of simple harmonic oscillator, Two body harmonic oscillator, An-harmonic oscillations, Motion in a frictional or drag environment, Algorithm development and problem solving	7	1,2,3,4

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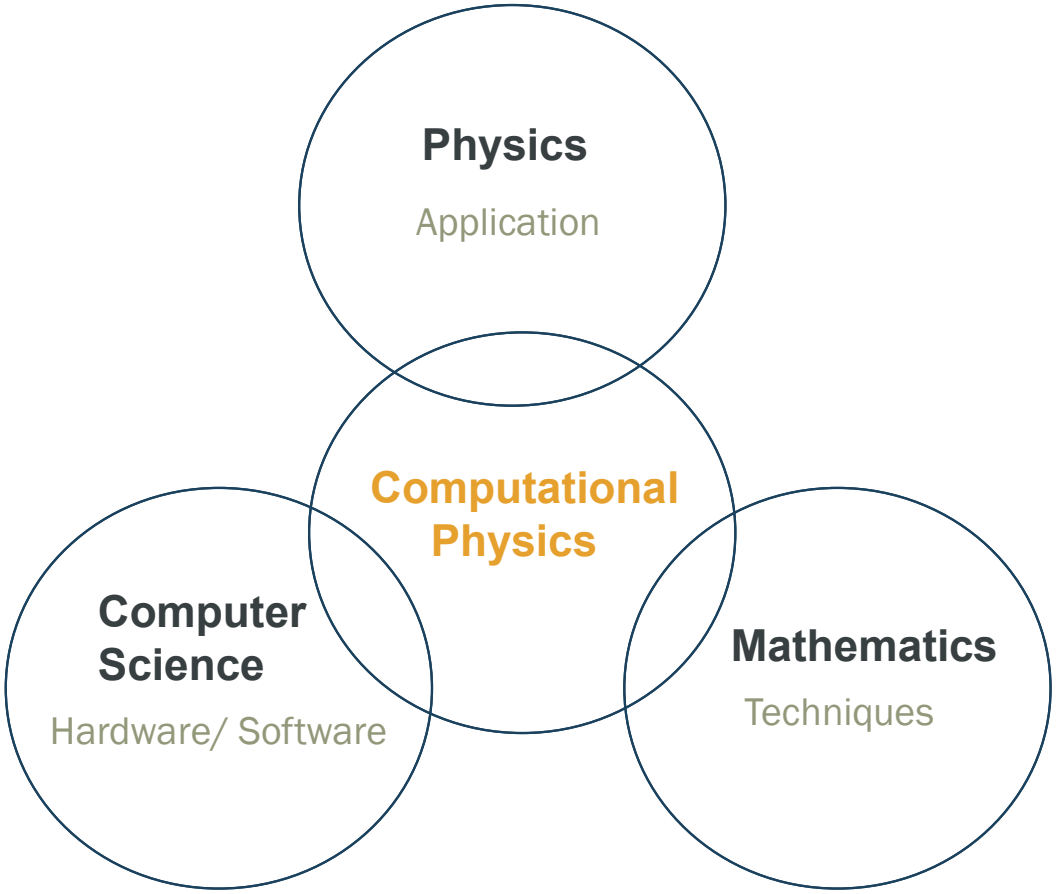
Computational Physics: what?



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Why computation ?

Finding Area of shapes analytically may not be very easy every time.

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Solving Real-World Problems:

- Free fall under gravity:
  - Not so free(Air Resistance)
- Simple Harmonic motion.
  - What if amplitude is large, Interactions,
- Hydrogen Atom(Only solvable problem in QM)
  - Solvable only if we consider Spinless, He atom not solvable.
- Sun-Earth system:
  - If we do not ignore moon?

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Saves effort, Money and time:

- Computations need less resource and time than real experiments.
- Work as a guide to Experiments by narrowing down.
- Easy to perform

Example: Designing a material for solar cell(ab-initio methods)
Evaluate battery performance(comsol)

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Understanding Physics:

- More control over computations allows us to understand the Interactions and Role of individual components.
- This understanding can leads to better designing.

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Need of Different approach:

- Scientific problems are often too complex to solve analytically.
- They need help and different approach for solution.
- Computers comes handy.

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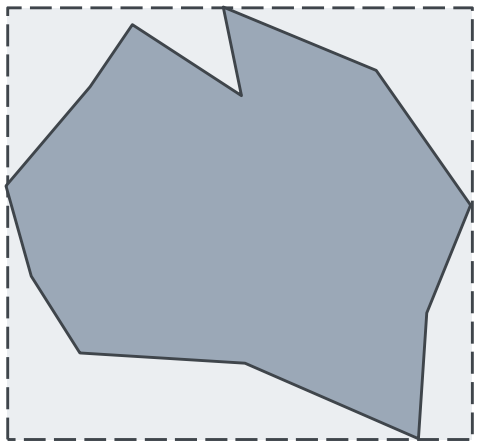
Example approach:



Step1: Draw a square around the shape.

Step2: Put random points in the square.

Step3: Count the number of points which falls inside the given area.

Step4: Ratio of points inside to total points.

We must take sufficient number of points for good estimation of area.

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Plots and Graphs

WHEN TO USE THEM?

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MOTIVE: *A picture is worth a thousand words.*

1. What are different kind Plots?
2. When to use them?
3. Qualities of good plot:
   - Less is more.

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Classical Graphs

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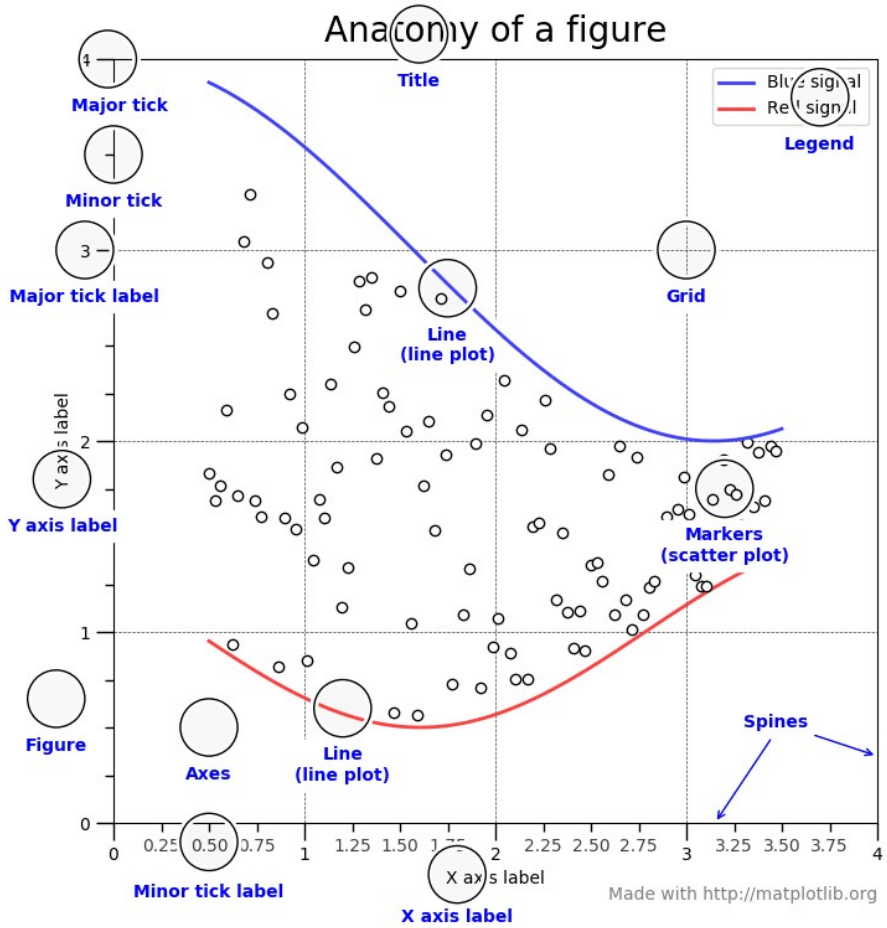

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Important Questions

What data I have?

- Quantitative
- Qualitative

What do I want to show?

- Relations
- Distribution
- Proportion
- Ranking

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Line plot

Area plot

Stacked area plot

Line plots/Area plots(Evolution)

- It shows connection between two points
- Deals with Quantitative data
- Points are ordered
- Shows evolution of variable
- Stacked plots good for showing cumulative as well as individual compare the different

Example: Time series data,
Website Traffic Analysis.
Global Temperature

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Bar plot

Horizontal bar plot

Bar plots(Ranking):

- Link between Qual. and Quant. Variables
- One bar = One category
- Easy to read
- The order is important.
- Horizontal plots are even easier to read

And helpful if category names are long .

Example: Subject wise Scores Comparison,
Product wise sales.

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Hist

Boxplot

Histogram and Boxplots(Distribution)

- Distribution of a Quant. Variable.
- How data distributed across different ranges.
- Height of each bar = Number of observations.
- Box plot gives summary of the data set like mean, deviation, range.

Example: Class Age Distribution,
Normal distribution(most common).

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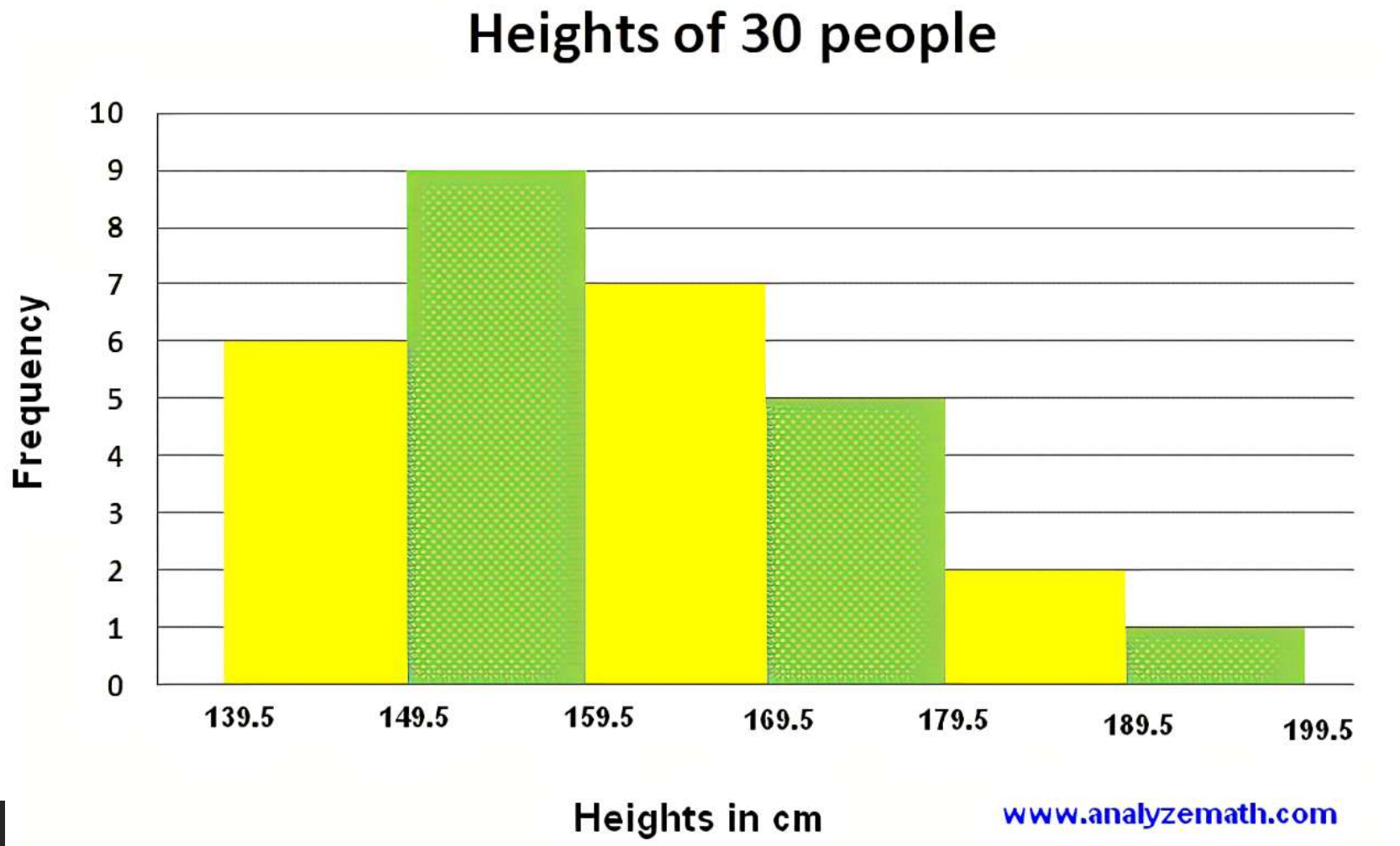

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Scatter plot

Heatmap

Scatter plot and Heatmap(Relations)

- Relation between two Quant. Variables.
- It can be used for Classification/Clustering problem

Example: Class Age Distribution

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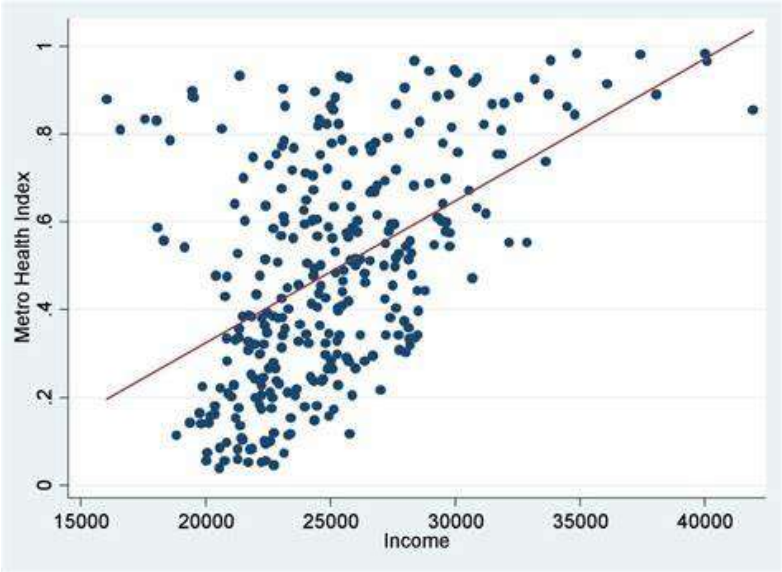

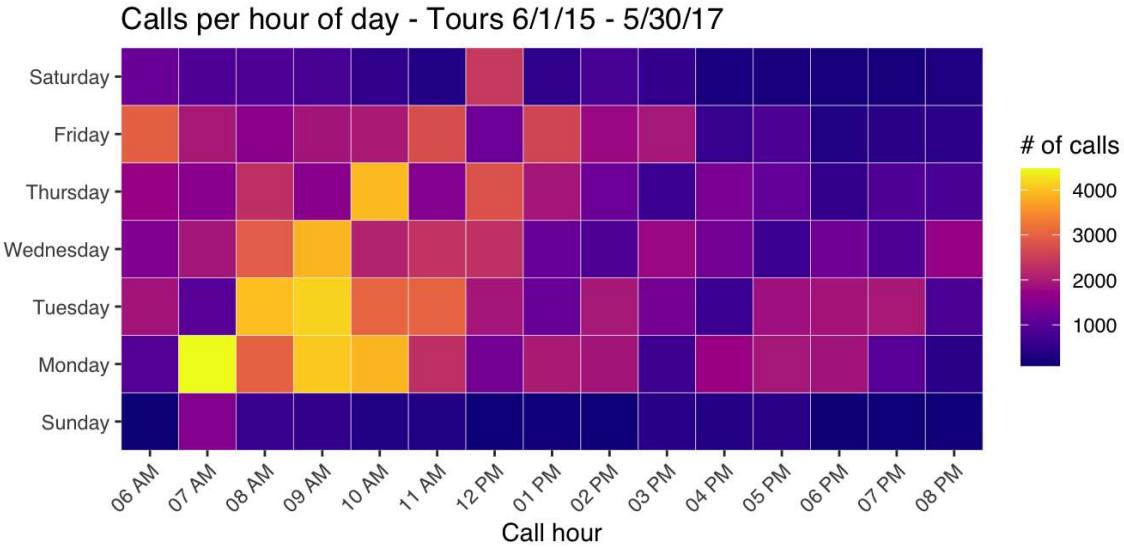

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Pie plot

Stacked bar plot

Pie plot and stacked Bar(Proportions)

The sum must be equal to 100%

- It can be used for Classification/Clustering problem

Example: Energy expanse.

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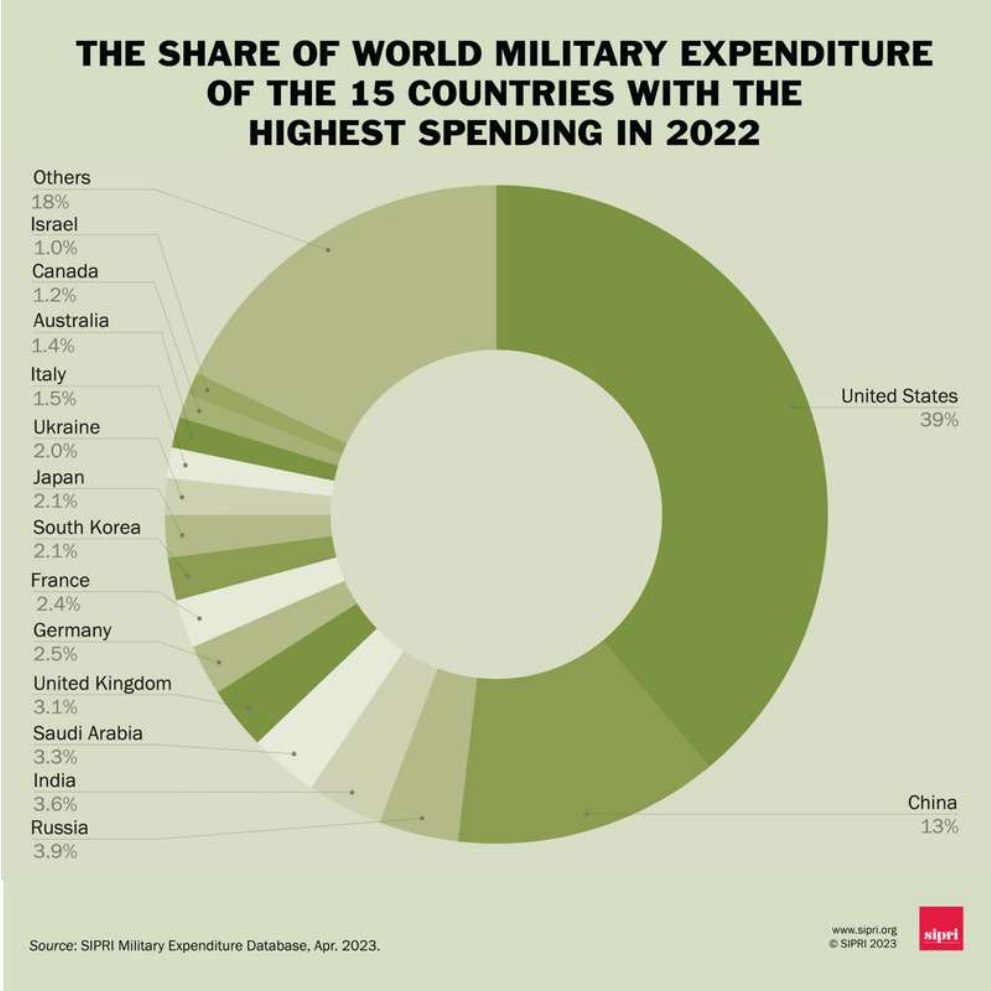

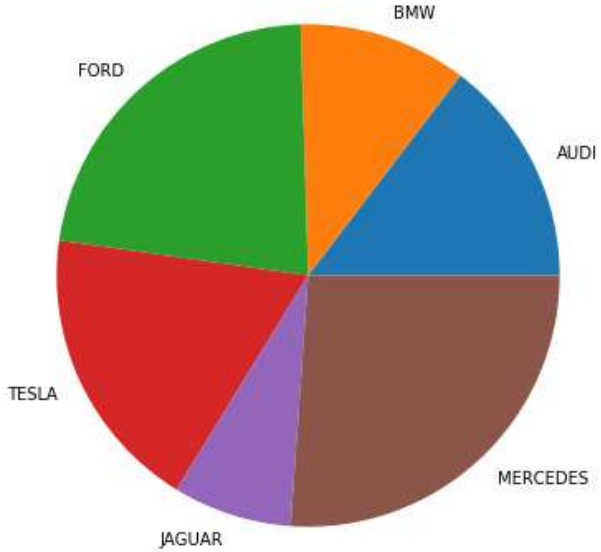

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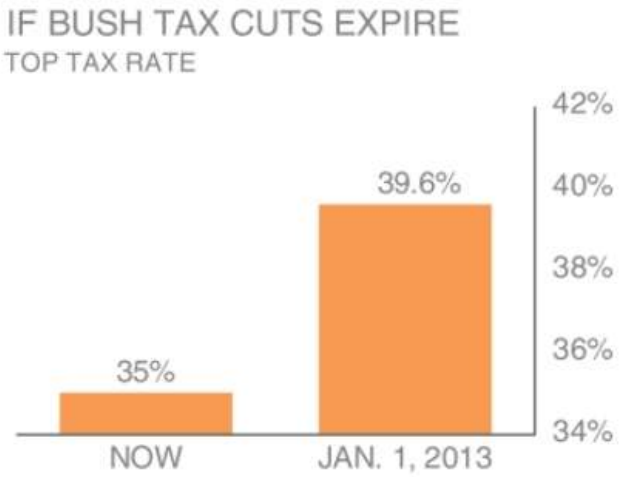
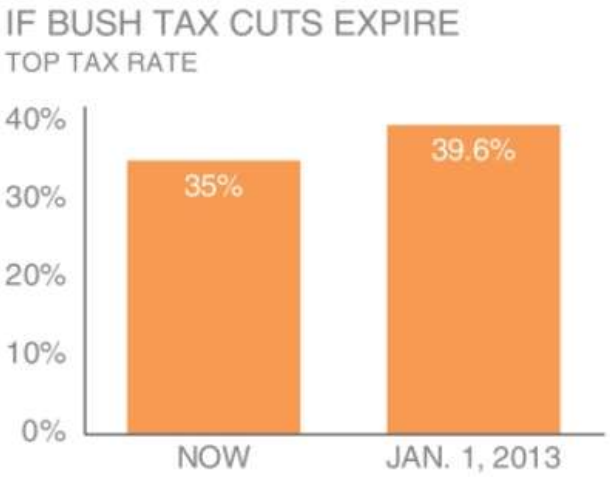
Examples of good plots:

Bar graph: baseline

Non-zero baseline: as originally graphed



Zero baseline: as it should be graphed



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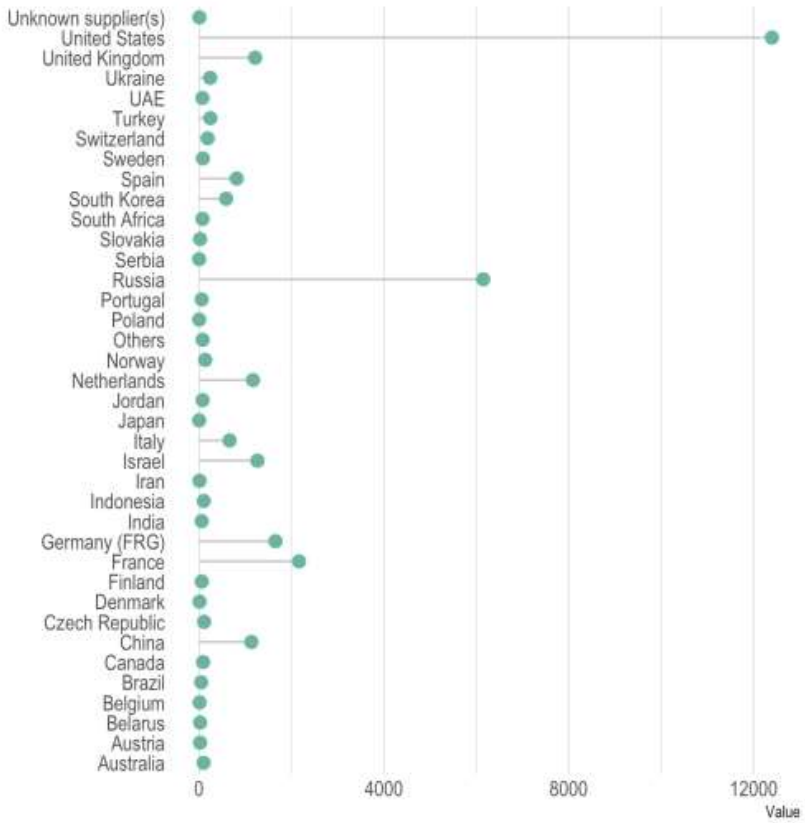
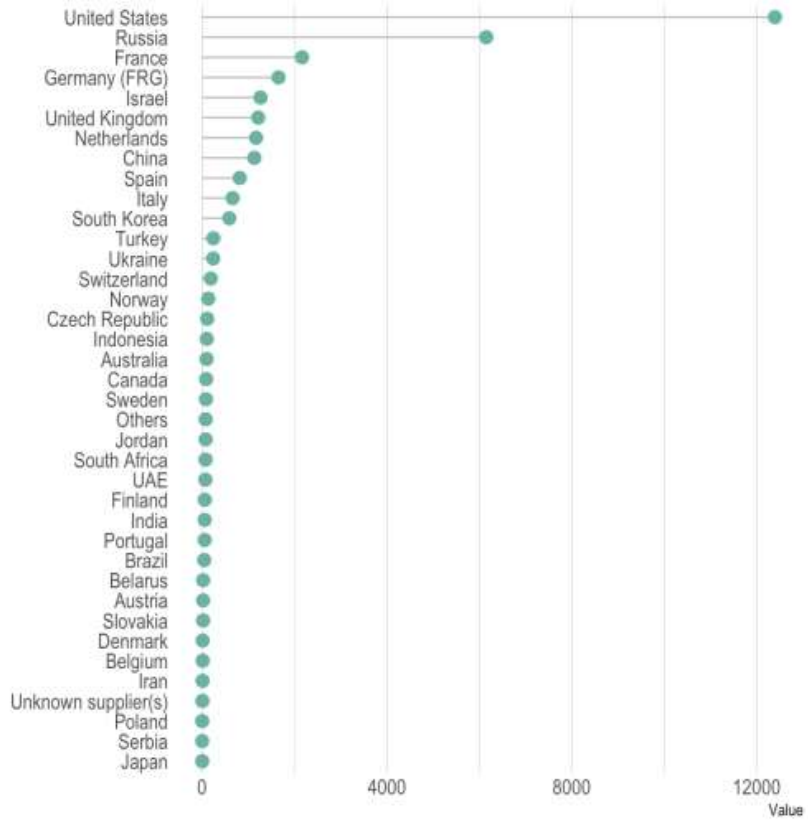
Bar graph : order your data





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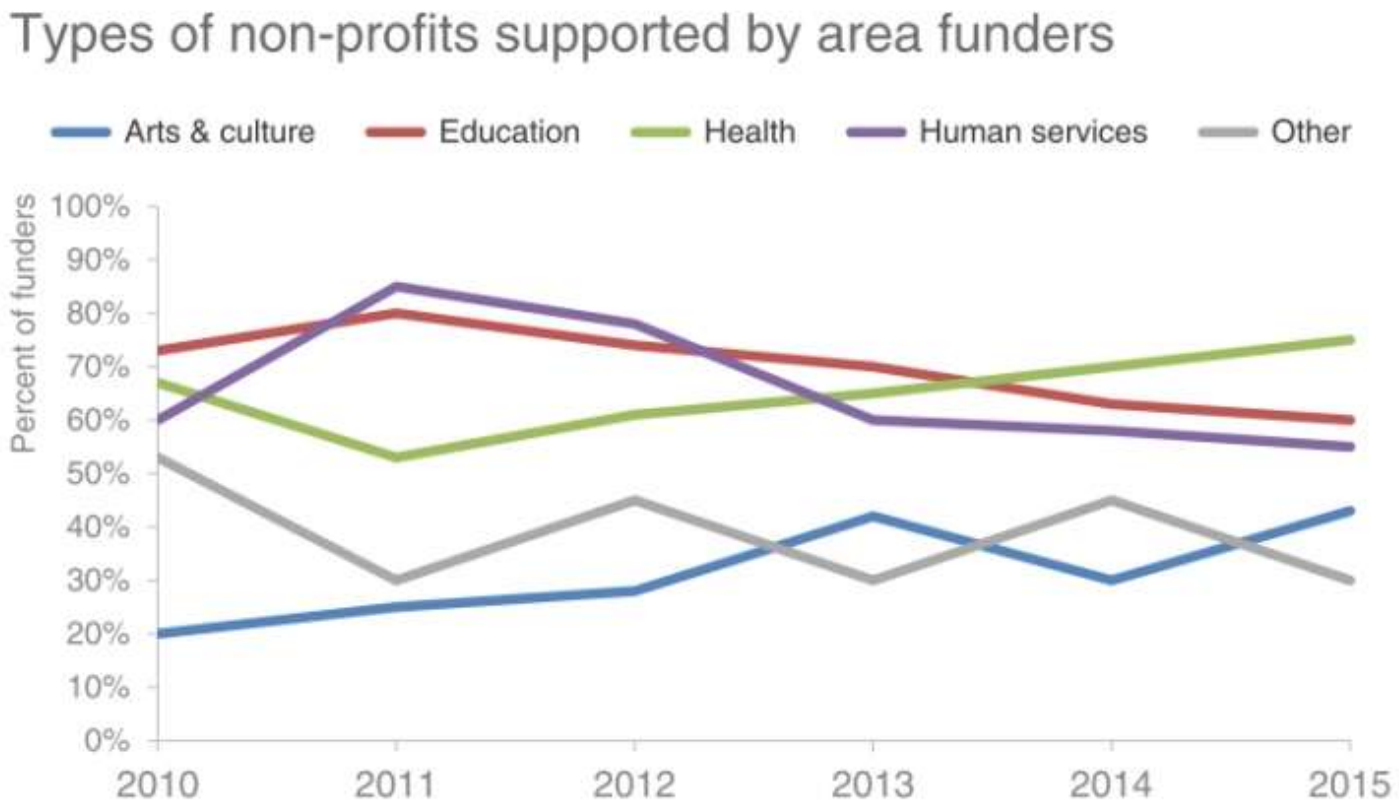

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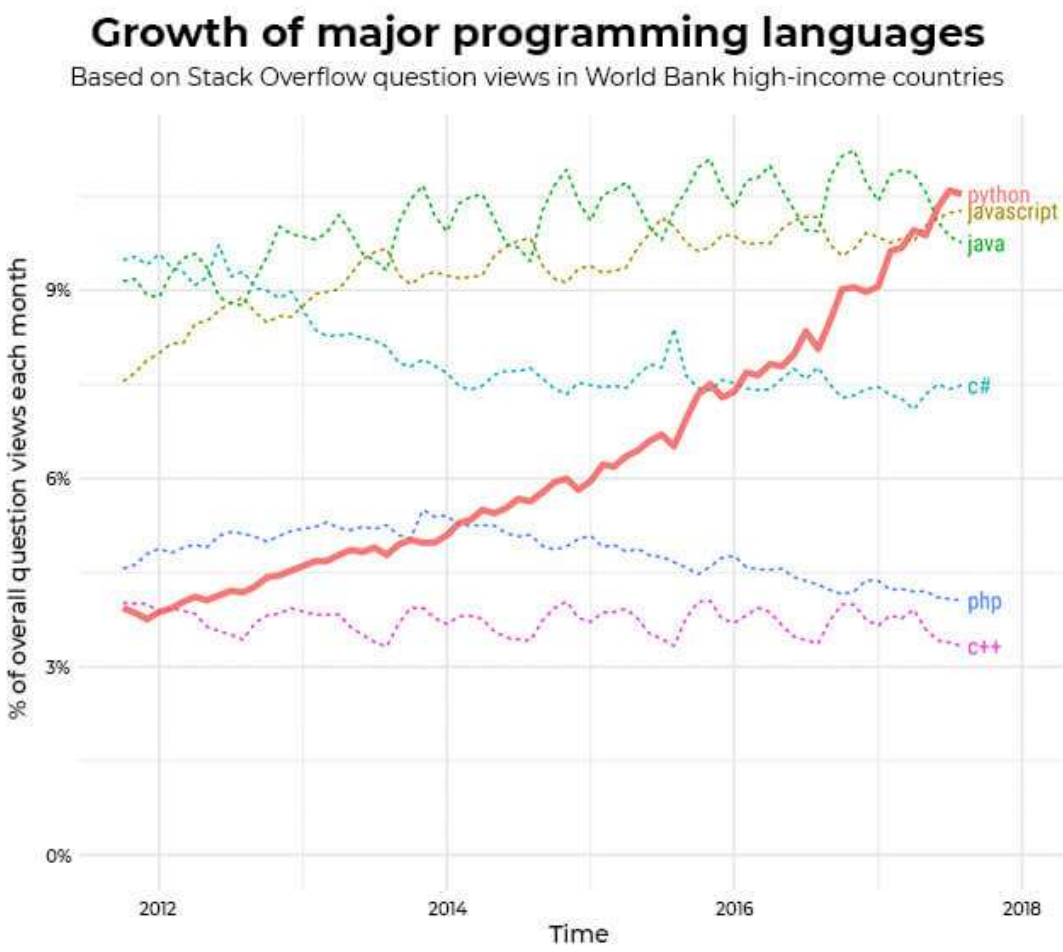

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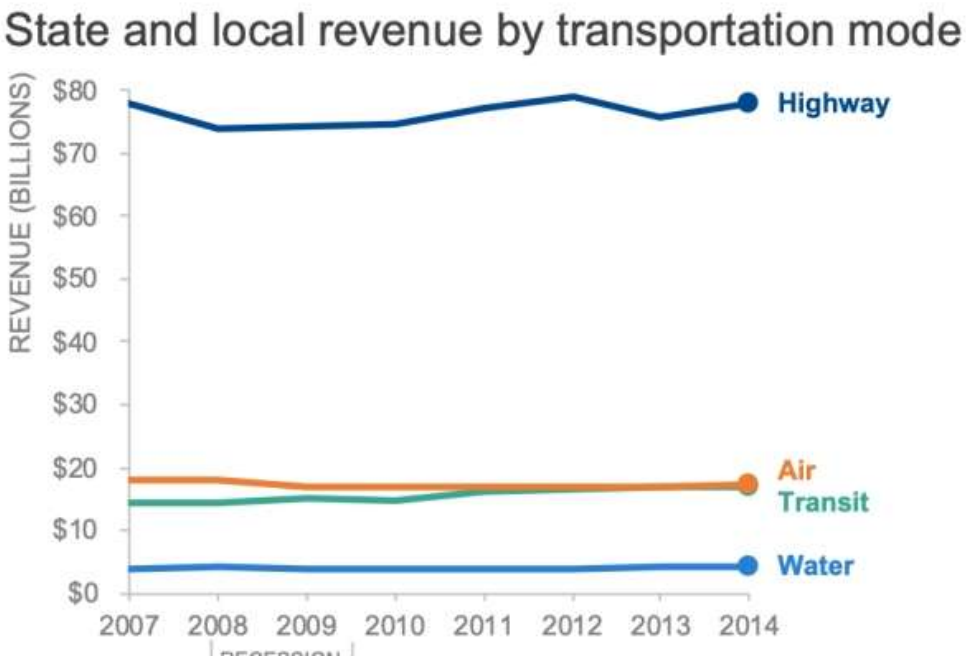
Line graph: range



Common challenge:

When one data series is so large relatively to the others that a single scale makes it nearly impossible to see any details.

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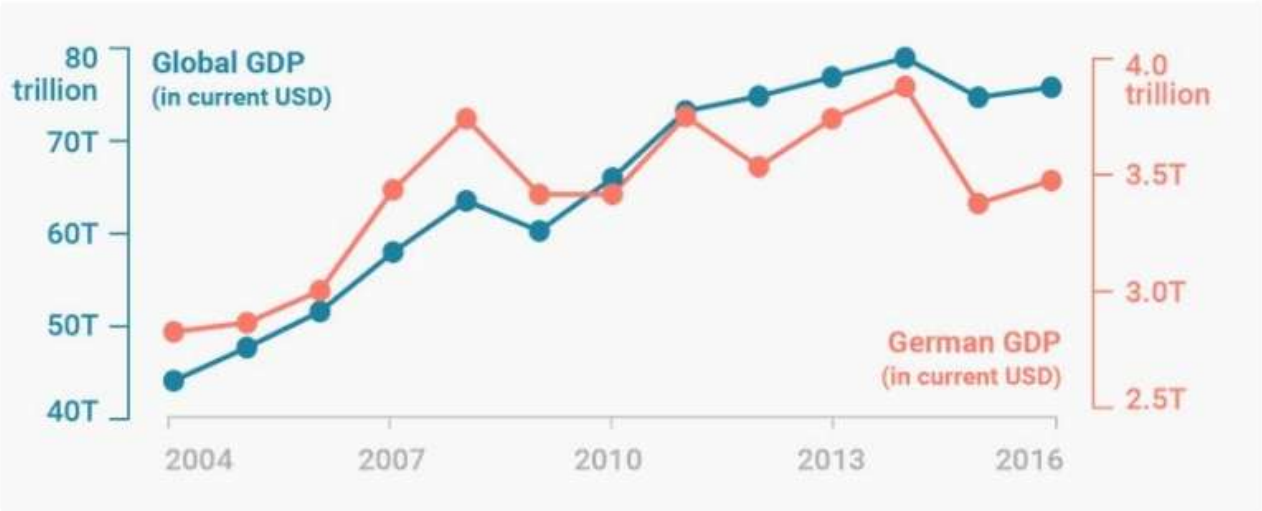
Line graph: dual axis



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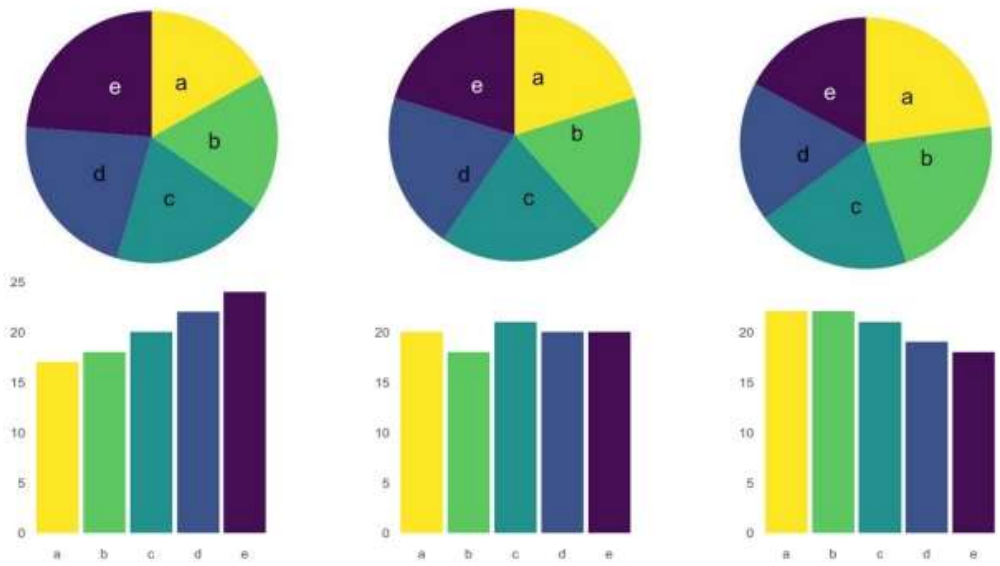
It's a different story when you want to compare similar proportions

The human eye is not very efficient at comparing angles It is much more efficient at comparing bar heights



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Line plot: to emphasize changes in values for one variable (plotted on the vertical axis) for continuous values of a second variable (plotted on the horizontal).

Multiple lines can also be plotted in a single line chart to compare the trend between series.

use a histogram to depict the frequency distribution of a single numeric variable. 2 not possible. histogram's primary variable is continuous and numeric.

Bar Chart: to show a distribution of data points or perform a comparison of metric values across different subgroups of your data. Distinct variable/ categorical.

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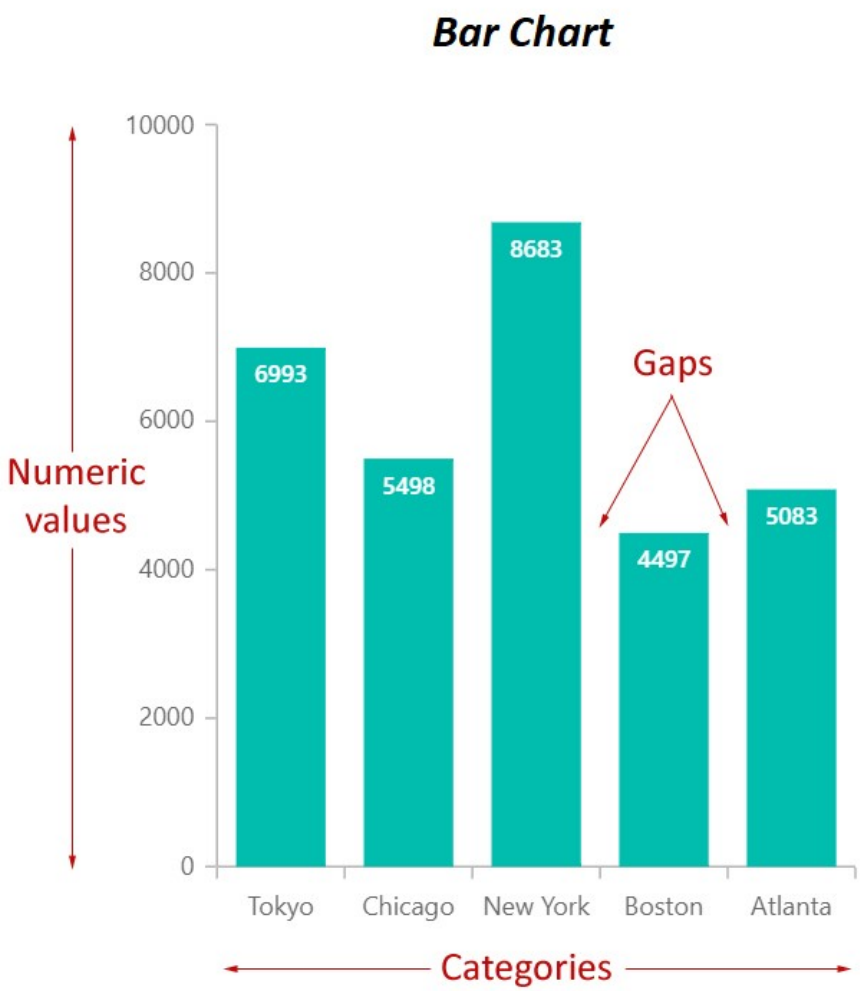

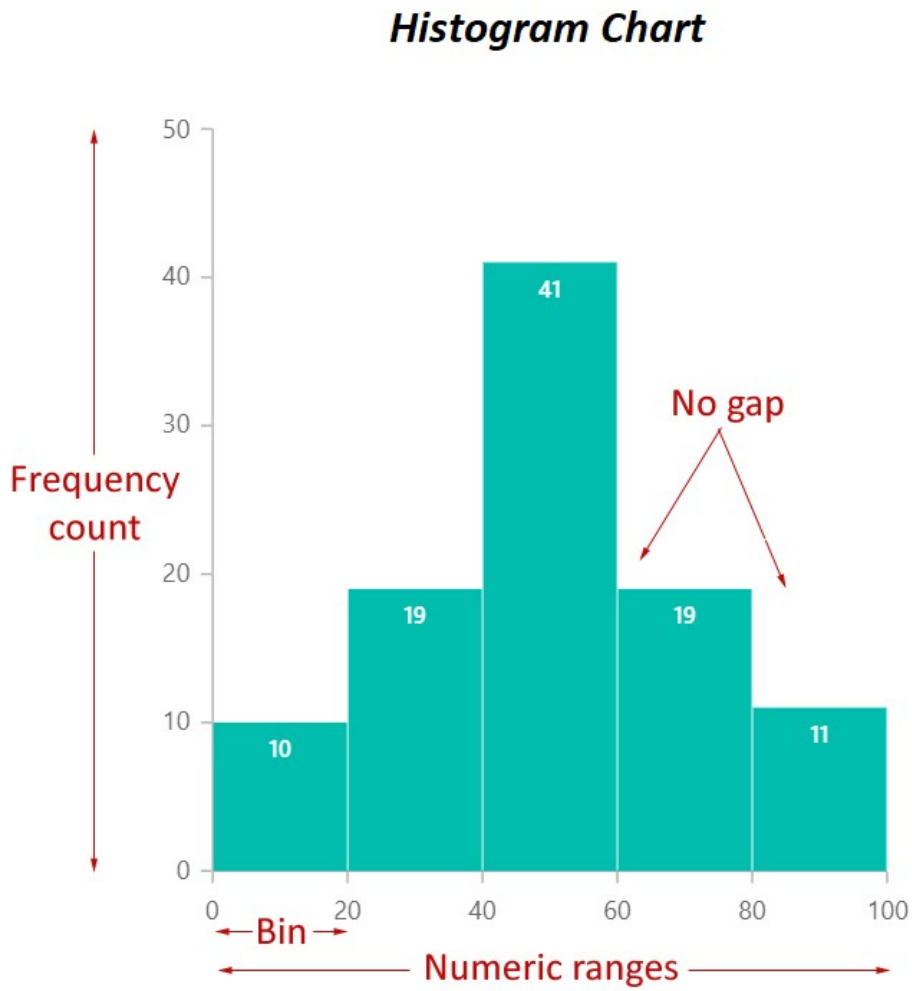

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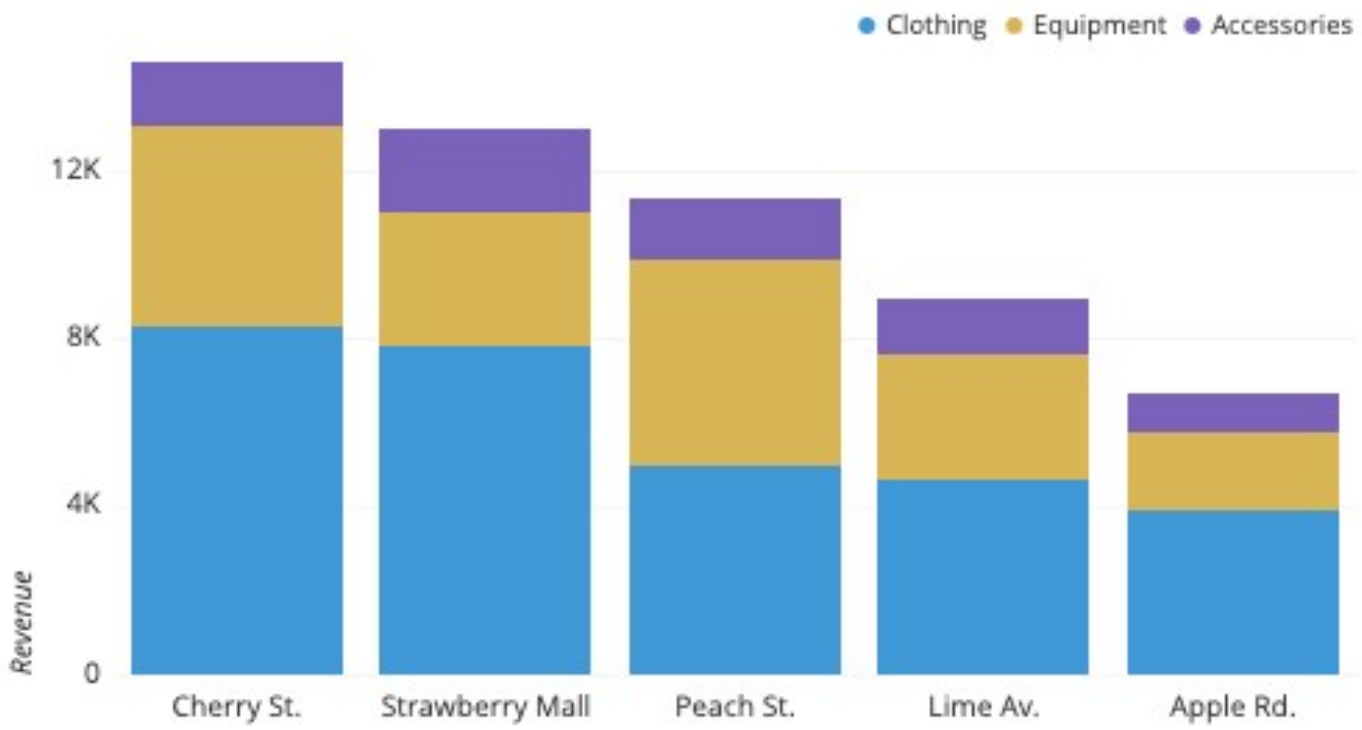

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Boxplot: to show distributions of numeric data values, especially when you want to compare them between multiple groups. High level info at a glance, data's symmetry, skew, variance, and outliers.

Stacked Bar:
relative decomposition of each primary bar based on the levels of a second categorical variable

Scatter plots' primary uses are to observe and show relationships between two numeric variables.
When over plotting use **heat maps**.

Pie and bar: Qualitative
Rest: Quantitative

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Reference:

https://www.data-to-viz.com

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Data interpretation:

- Giving meaning to the data drawing conclusions, and understanding relationships or patterns within the data.
- Depending on data type, there are two main methods of data interpretation are:
  - Qualitative
  - Quantitative

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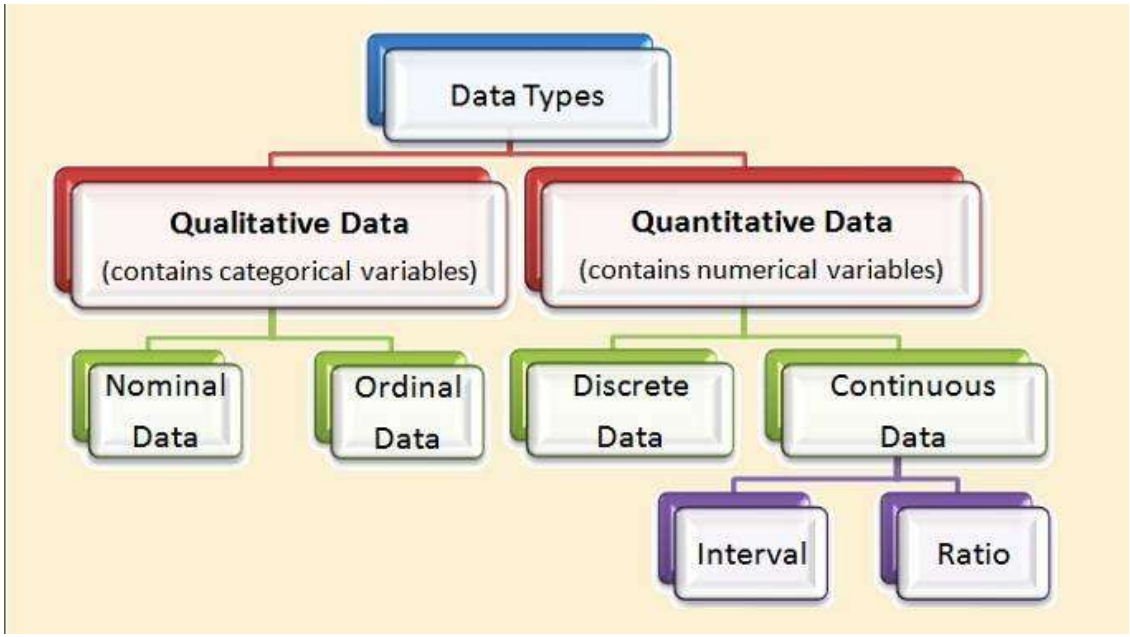

Qualitative data can take on numerical values, but those numbers have little mathematical meaning.

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Nominal Data:

- Nominal values are Labels represent discrete units,
- No quantitative value.
- nominal data that has no order.

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Are you married?

- Yes
- No

What languages do you speak?

- Englisch
- French
- German
- Spanish

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Ordinal Data:

- Ordinal values are values represent discrete and ordered units.
- No Quantitative significance: **Arithmetic operations** (e.g., adding or averaging) can not be performed on such categorical data.
- Ordering matters.

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What Is Your Educational Background?

- 1 - Elementary
- 2 - High School
- 3 - Undegraduate
- 4 - Graduate

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Quantitative Data:

Discrete Data:

- We speak of discrete data if the data can only take on certain values.
- Can you count it and can it be divided up into smaller and smaller parts? Ordering matters.

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Examples :

1. Number of students in a class, can't have 20.5 students.

2. Number of heads in a series of coin flips.

3. Number of pets owned by a family.

In computational physics, the majority of data tends to be **quantitative** and **continuous**, since scientific measurements often involve precise values that can vary infinitely, such as time, temperature, or distance.

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Interval Data:

Definition: Interval data is measured on a scale where the intervals between each value are equal. However, there is **no true zero point**, meaning zero does not represent the absence of the quantity.

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Examples:

- Temperature in Celsius or Fahrenheit (0°C doesn't mean "no temperature," it's just a point on the scale).
- Calendar years (the year 0 doesn't mean time didn't exist).

Key Characteristics:

- Values can be positive or negative.
- Arithmetic operations like addition and subtraction can be performed.

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Ratio Data:

Definition: Ratio data also has equal intervals between points, but unlike interval data, it has a **true zero point**, representing the absence of the quantity being measured.

Examples:

- Weight (0 kg means no weight).
- Height (0 cm means no height).
- Distance (0 meters means no distance).

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Key Characteristics:

- Zero represents the complete absence of the characteristic.
- All mathematical operations, including multiplication and division, can be performed.
- No negative values exist in ratio data.

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Making Predictions

If we know temperature at 10 AM and 12 PM.

What would be Temperature at 11 AM ?

If we know score in 14th over, what would be the score at the end of 20th over?

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Making Predictions(Interpolation and Extrapolation):

- How to predict?
- **Within the range** of a discrete set of known data points (Interpolation)
- Outside the range(Extrapolation).
- If we know temperature at 10 AM and 12 PM, interpolation allows you to estimate the temperature at 11 AM. Temperature at 1PM is extrapolation.

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How to do it??

- **Interpolation/Extrapolation** relies on having a function.

- **Curve Fitting** is the process where we try to associate a function with the data by adjusting the parameters. The function should be the best function that can explain the data.

- Once the curve (or function) is established, we can use **interpolation** and **extrapolation**.

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Curve Fitting:

Curve fitting is to find a mathematical function that best represents the relation between the independent and dependent variables in a dataset.

The goal is to minimize the differences between the predicted values (from the curve) and the actual data points.

Least Squares Method: The most common approach to curve fitting, which minimizes the sum of the squares of the differences between the data points and the curve.

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Steps in Polynomial Fitting:

1. **Define the Polynomial**: For a given degree n, the polynomial has the general form:

$$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

   where $a_0, a_1, a_2, \ldots, a_n$ are the coefficients to be determined.

2. **Data Points**: Given a set of data points $(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)$, you want to find the polynomial that approximates these points.

3. **Minimization of Error**: Typically, the **least-squares method** is used. The idea is to minimize the sum of the squares of the residuals, which are the differences between the observed data points y_i and the values predicted by the polynomial $P(x_i)$:

$$\text{Error} = \sum_{i=1}^{m} (y_i - P(x_i))^2$$

   By minimizing this error, you find the opt coefficients for the polynomial.

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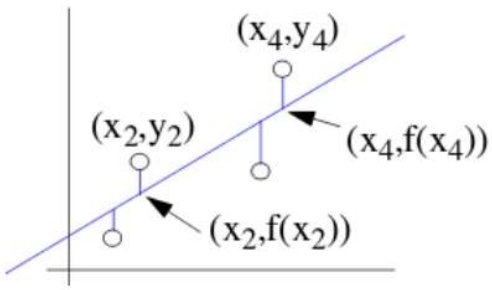
Linear Fitting Using Least Squares

The linear model is of the form y=ax+b.

We need to **minimize the sum of squared errors.**

$$S = \sum_{i=1}^{n} (y_i - (ax_i + b))^2$$



Where n is the number of data points.

Minimization Conditions:

$$\frac{\partial s}{\partial a} = 0 \qquad \frac{\partial s}{\partial b} = 0$$

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Equations for the least squares fit:

$$\sum_{i=1}^{n} y_i = a \sum_{i=1}^{n} x_i + b_n$$

$$\sum_{i=1}^{n} x_i y_i = a \sum_{i=1}^{n} x_i^2 + b \sum_{i=1}^{n} x_i$$

In Matrix form:

$$\mathrm{B} = \boldsymbol{A}.\boldsymbol{X} \qquad A = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}, \quad X = \begin{bmatrix} b \\ a \end{bmatrix}, \quad B = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \end{bmatrix}$$

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Linear Fit

Linear equation: $$y = ax + b$$

$$S = \sum_{i=1}^{n} (y_i - (ax_i + b))^2$$

Minimization Conditions:

$$\frac{\partial s}{\partial a} = 0 \qquad \frac{\partial s}{\partial b} = 0$$

Leads to:

$$\sum_{i=1}^{n} y_i = a \sum_{i=1}^{n} x_i + b_n$$

$$\sum_{i=1}^{n} x_i y_i = a \sum_{i=1}^{n} x_i^2 + b \sum_{i=1}^{n} x_i$$

Quadratic Fit

Quadratic equation: $$y = ax^2 + bx + c$$

$$S = \sum_{i=1}^{n} (y_i - (ax_i^2 + bx_i + c))^2$$

Minimization Conditions:

$$\frac{\partial s}{\partial a} = 0 \qquad \frac{\partial s}{\partial b} = 0 \qquad \frac{\partial s}{\partial c} = 0$$

Leads to:

$$\sum_{i=1}^{n} y_i = a \sum_{i=1}^{n} x_i^2 + \mathrm{b} \sum_{i=1}^{n} x_i + cn$$

$$\sum_{i=1}^{n} x_i y_i = a \sum_{i=1}^{n} x_i^3 + \mathrm{b} \sum_{i=1}^{n} x_i^2 + c \sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} x_i^2 y_i = a \sum_{i=1}^{n} x_i^4 + \mathrm{b} \sum_{i=1}^{n} x_i^3 + \mathrm{c} \sum_{i=1}^{n} x_i^2$$

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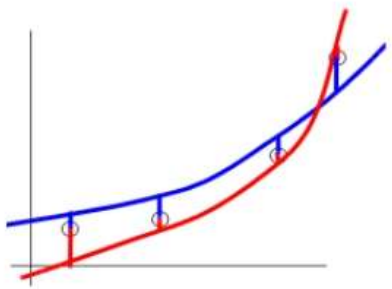
Curve fitting



Polynomial fitting

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_j x^j = a_0 + \sum_{k=1}^{j} a_k x^k$$

$$a_0, a_1, a_2, a_3, \ldots\ldots\ldots, a_j \,?$$

$$err = \sum (d_i)^2 = (y_1 - f(x_1))^2 + (y_2 - f(x_2))^2 + (y_3 - f(x_3))^2 + (y_4 - f(x_4))^2$$

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_j x^j = a_0 + \sum_{k=1}^{j} a_k x^k$$

$$err = \sum_{i=1}^{n} \left(y_i - \left(a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3 + \ldots + a_j x_i^j \right) \right)^2$$

$$err = \sum_{i=1}^{n} \left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k \right) \right)^2$$

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Curve fitting

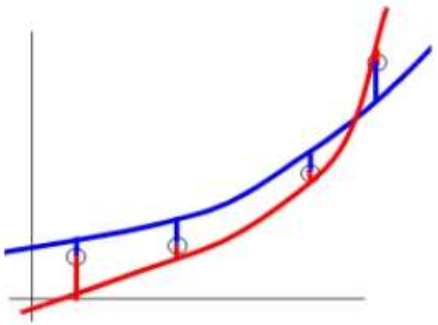
$$\frac{\partial err}{\partial a_0} = -2\sum_{i=1}^{n}\left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k\right)\right) = 0$$

$$\frac{\partial err}{\partial a_1} = -2\sum_{i=1}^{n}\left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k\right)\right) x = 0$$

$$\frac{\partial err}{\partial a_2} = -2\sum_{i=1}^{n}\left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k\right)\right) x^2 = 0$$

$$\vdots$$

$$\frac{\partial err}{\partial a_j} = -2\sum_{i=1}^{n}\left(y_i - \left(a_0 + \sum_{k=1}^{j} a_k x^k\right)\right) x^j = 0$$



$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 & \dots & \sum x_i^j \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \dots & \sum x_i^{j+1} \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 & \dots & \sum x_i^{j+2} \\ \vdots & \vdots & \vdots & & \vdots \\ \sum x_i^j & \sum x_i^{j+1} & \sum x_i^{j+2} & \dots & \sum x_i^{j+j} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_j \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \\ \sum \left(x_i^2 y_i\right) \\ \vdots \\ \sum \left(x_i^j y_i\right) \end{bmatrix}$$

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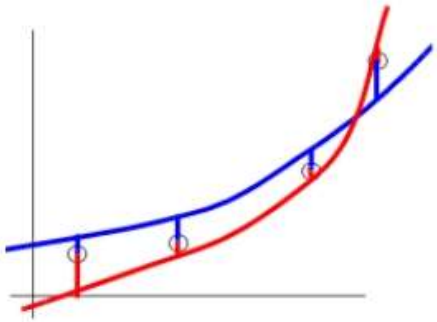
Curve fitting

$$A = \begin{bmatrix} n & \sum x_i & \sum x_i^2 & \dots & \sum x_i^j \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \dots & \sum x_i^{j+1} \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 & \dots & \sum x_i^{j+2} \\ \vdots & \vdots & \vdots & & \vdots \\ \sum x_i^j & \sum x_i^{j+1} & \sum x_i^{j+2} & \dots & \sum x_i^{j+j} \end{bmatrix}, \quad X = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_j \end{bmatrix}, \quad B = \begin{bmatrix} \sum y_i \\ \sum (x_i y_i) \\ \sum \left(x_i^2 y_i\right) \\ \vdots \\ \sum \left(x_i^j y_i\right) \end{bmatrix}$$



$$AX = B$$

$$X = A^{-1} * B$$

$$na_0 + \Sigma x_i a1 + \Sigma x_i^2 a_2 = \Sigma y_i$$
$$\Sigma X_i a_0 + \Sigma x_i^2 a_1 + \Sigma x_i^3 a_2 = \Sigma x_i y_i$$
$$\Sigma X_i^2 a_0 + \Sigma x_i^3 a_1 + \Sigma x_i^4 a_2 = \Sigma x_i^2 y_{ix}$$

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Biasing:

- Biasing refers to systematic error or deviation from the true value or reality. It distort the conclusions making models less accurate and less reliable.
- **Sample Bias** - Sample collected for a study is not representative of the whole Example: Collecting income data from Ambanis and Adani for surveying average income of Indians.
- **Measurement Bias** - Whan the method of measurement systematically differs between groups. - Example: Inaccurate weight scales in a study.

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- **Observer Bias:** When the researcher unconsciously influences the experiment or data collection due to subjective expectations or interpretations. - Example: A researcher expecting a particular outcome might record results in a way that favors that outcome.
- **Confirmation Bias** - The tendency to focus on or give more weight to evidence that supports one's pre-existing beliefs, while ignoring evidence that contradicts them. - Example: Reading studies that support your hypothesis while disregarding those that do not.

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Reducing Bias in Data collection:

Drastically affect the validity and reliability of data and research.

- **Randomize sampling** to avoid selection bias.
- **Design objective surveys** and measurement tools to prevent response and measurement biases.
- **Double-blind studies** can help mitigate observer bias.
- **Report all findings**, including negative or null results, to avoid publication bias.

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Exercises: "Which Plot Should You Use?"

1. A company records its **monthly revenue for 5 years**.
2. The **hourly temperature** in your city for a week is collected.
3. The **daily rainfall (in mm)** during monsoon season is measured.
4. The **weekly closing price of Nifty50** is recorded for 2 years.

5. The **ages of 500 people** in a town are collected.
6. A teacher collects **math scores of 100 students** and wants to see how they are distributed.
7. A researcher measures the **time (in minutes)** people spend exercising daily.
8. A scientist records **particle sizes** in a material sample.

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9. Survey of favorite **fast food items**: Pizza, Burger, Sandwich, Fries.
10. A company compares the **sales of four products (A, B, C, D)** in one year.
11. A library records how many books were borrowed in categories: **Fiction, Non-Fiction, Science, History, Kids**.
12. A teacher compares **boys vs. girls average marks** in three subjects.

13. Smartphone **market share**: Apple, Samsung, OnePlus, Xiaomi, Others.
14. A pie chart of **budget spending**: Rent, Food, Travel, Savings, Entertainment.
15. Distribution of **languages spoken** in a city: Hindi, English, Marathi, Bengali, Others.

16. **Height vs. Weight** of 100 people.
17. **Study time vs. Exam score** for 50 students.
18. **Temperature vs. Ice cream sales** in summer.
19. **Advertising spend vs. Sales revenue** for 20 companies.

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20. Exam marks of **Class A and Class B** in Science.
21. Monthly salaries of employees in **two companies**.
22. Blood pressure readings for **men vs. women** in a study.
23. The **reaction times** of people under 3 different conditions: rested, tired, caffeinated.

24. Checking **household electricity bills** for unusual spikes.
25. Distribution of **athletes' race completion times** to spot very slow/fast outliers.
26. Comparing **monthly expenses** of families to see which have unusual spending patterns.
27. Checking **stock returns** of companies to identify risky ones.

28. A student tracks **steps walked daily** for 1 month and wants to see improvement.

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29. A hospital records the **number of patients per department** in a single day.
30. A researcher wants to check whether **two measurements (length vs. width of leaves)** are linearly related.
31. A political survey shows **percentage vote share** of different parties in an election.
32. A teacher wants to quickly compare the **highest-scoring subject** in a class.
33. An economist wants to see the **trend of inflation rate over 10 years**.
34. A doctor wants to check the **spread of blood sugar levels** among 100 patients.
35. A traffic survey records the **count of cars, bikes, buses, and trucks** crossing a signal in one hour.

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Previous Year Questions from Module-1

1	(a)	Illustrate the following i) Nominal data ii) Ordinal data iii) Interval data iv) Ratio data	6
	(b)	Explain the methods to convert a high level programming computer program into machine-readable format. Explain characteristics, advantage and disadvantage in each case.	6
		OR	
	(c)	Create a flowchart to detailing the process of the curve fitting using a nth order polynomial function.	12

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6		Fit the given data using the second order polynomial $y=a_0+a_1x+a_2x^2$	8

x	1.0	2.0	3.0	4.0
y	6.0	11.0	18.0	27.0

1	(a)	With the help of appropriate example (include the flow chart for the example) explain flow control structures in the programming language.	**7**
1	(b)	A company conducts a survey to rate a product based upon the price and quality. Classify the data thus obtained and methods to analyse it.	**3**

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2

x	0	0.5	1.0	1.5	2.0	2.5
y	-0.4326	-0.1656	3.1253	4.7877	4.8535	8.6909

Evaluate the function f(x) for a linear fit of the above given data. **10**

(a)

X	0	1	2	3	4	5	6	7	8
Y	12.0	10.5	10.0	8.0	7.0	8.0	7.5	8.5	9.0

Obtain the best fit for the above given data using the second order polynomial. 12

(b) What do you understand by term 'bias'. Describe the different ways bias can be introduce in data/conclusion. Also, explain the effect of bias on outcome of data analysis. 12

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6		Illustrate the following with the help of a suitable example 1. Nominal data 2. Ordinal data Also, discuss how would you analyze the above given data types.	8

6		Discuss the following and provide the utilization of each 1. Bar graph 2. Histogram 3. Pi chart	8

1	(a)	<table><tr><td>X</td><td>1</td><td>6</td><td>11</td><td>16</td><td>20</td><td>26</td></tr><tr><td>Y</td><td>13</td><td>16</td><td>17</td><td>23</td><td>24</td><td>31</td></tr></table> Obtain the best linear fit for the above given data.	12
		OR	
	(b)	Derive the generalized solution for curve fitting problem using a nth order polynomial.	12

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